Modeling and Controller Design of Distribution Static Synchronous Compensator

Yubin Wang, Jiwen Li, Yan Lv and Xuelian Liu

***Abstract*--Distribution Static Synchronous Compensator (DSTATCOM) is an effective measure to maintain voltage stability and improve power quality of distribution grid. This paper deals with the modeling and nonlinear controller design of DSTATCOM. A nonlinear dynamic mathematical model of DSTATCOM is obtained by the switching function approach. A reactive-voltage decoupled PWM controller is designed based on multivariable inverse-system method. For the decoupled system, a control strategy combining control of state feedback and feedforward is employed, which can make the system have good transient response performance and anti-disturbing ability. By this method, the controller parameters' design of both reactive current and dc-link voltage are simplified effectively. The theoretical analysis and design are verified by the simulation results.**

***Index Terms*—DSTATCOM, Dynamic Modeling, Controller design, Inverse-system method, Reactive-voltage decoupled control**

I. INTRODUCTION

AS an important application of power electronics in power system, the technology of Static VAR Compensation has been developing quickly in recent years. Distribution Static Synchronous Compensator (DSTATCOM), one of the second generation DFACTS, is an effective measure to maintain voltage stability and improve power quality of distribution grid.

DSTATCOM is a Voltage Source Converter (VSC) which can provide or absorb reactive power depending on the grid state. Considering the dissymmetry of power system and unbalance of loads, DSTATCOM is expected to be single-phase controllable. Therefore, the main circuit configuration is a three-phase inverter consisting of three Single-Phase Full-Bridge(SPFB) inverters. Then a dynamic mathematical model of DSTATCOM, whose main circuit consists of SPFB VSCs, is established by using the switching function approach in this paper.

But, whatever approach is used, the mathematical model will be nonlinear, this lies on DSTATCOM itself. It is difficult to design a controller for a nonlinear model, and the usual method is linearization of the nonlinear model for small perturbations[1]. A steady-state equilibrium point is specified in this method, but in fact the operation point is variable and unstable, so a controller with constant parameters based on this method is not robust enough. Therefore, a reactive current-voltage decoupled PWM controller is designed based on multivariable inverse-system method[2,10], and for the decoupled system, a control strategy combining control of state feedback and feedforward is employed, which can make system have good transient response performance and anti-disturbing ability. By this method, the controller parameters' design of both reactive current and dc-link voltage are simplified effectively. The theoretical analysis and design are verified by the simulation results.

II. MAIN CIRCUIT CONFIGURATION

Before modeling, the main circuit configuration of DSTATCOM must be chosen first. Usually, the following factors should be considered:

1) Because of the serious asymmetry of distribution grid, DSTATCOM is expected to be single-phase controllable.

2) The DSTATCOM has advantages of low cost, good economy, as well as high reliability and easy controllability.

3) It is easy to carry out expansion of capacity and voltage level, easy to carry out module and redundancy design.

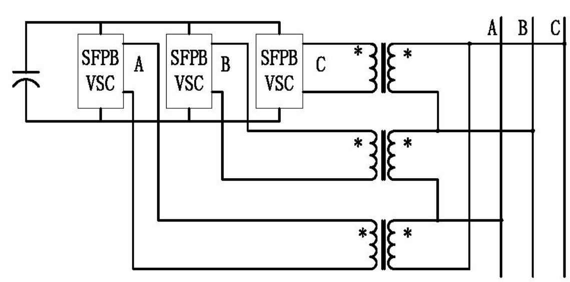

Fig. 1. Basic circuit configuration of DSTATCOM

Therefore, a three-phase VSC with tie-transformer consisting of three single-phase full-bridge (SFPB) inverters is a good choice of DSTATCOM's circuit configuration, and a schematic diagram is shown in Fig.1. It can carry out single-phase control easily; On the other hand, the dc sides of all three SFPB inverters share a common capacitor (shown in Fig.2), thus the problem of voltage unbalance on dc sides can be avoided, and the voltage fluctuation of dc-link capacitor can reduce more than 50%. Furthermore, it can not only increase the voltage level through the tie-transformer, but also carry out module and redundancy design through paralleling

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with new SFPB units in each original phase.

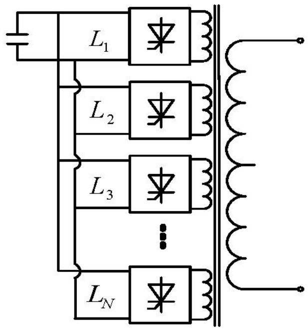


Fig. 2. The single-phase schematic diagram of all VSC dc sides sharing a common capacitor

III. DSTATCOM DYNAMIC MODELING

A single-phase equivalent circuit of DSTATCOM is shown in Fig.3. Where Rc is included to represent small losses in the switching devices of VSC. R_S and L represent the equivalent circuit of the tie-transformer between system voltage U_S and the output voltage U_I of DSTATCOM.

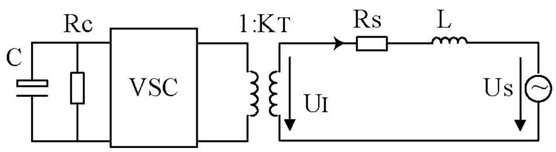


Fig. 3. The single-phase equivalent circuit of DSTATCOM

The u_{sa}, u_{sb}, u_{sc} are defined as instantaneous values of system phase voltage, and can be given by

$$\begin{cases} u_{sa} = \sqrt{2}U_S \sin \omega t \\ u_{sb} = \sqrt{2}U_S \sin(\omega t - \frac{2}{3}\pi) \\ u_{sc} = \sqrt{2}U_S \sin(\omega t + \frac{2}{3}\pi) \end{cases} \tag{1}$$

Where U_S is rms value of system phase voltage.

The output voltages of DSTATCOM, u_{ia}, u_{ib} and u_{ic} can be given by

$$\begin{cases} u_{ia} = \frac{\sqrt{3}}{3} K_T m U_{dc} \sin(\omega t + \delta) \\ u_{ib} = \frac{\sqrt{3}}{3} K_T m U_{dc} \sin(\omega t + \delta - \frac{2\pi}{3}) \\ u_{ic} = \frac{\sqrt{3}}{3} K_T m U_{dc} \sin(\omega t + \delta + \frac{2\pi}{3}) \end{cases} \tag{2}$$

Where K_T is turn's ratio of the tie-transformer, m is the amplitude modulation ratio of VSC' output voltage, its value depends on the type of VSC. U_{dc} is the dc-link capacitor's voltage of VSC, and δ is the phase angle difference between voltage u_s and u_i.

The following equations can be derived out by KVL and KCL from Fig.3:

$$\begin{cases} L\frac{di_a}{dt} = u_{sa} - u_{ia} - R_S i_a \\ L\frac{di_b}{dt} = u_{sb} - u_{ib} - R_S i_b \\ L\frac{di_c}{dt} = u_{sc} - u_{ic} - R_S i_c \\ CU_{dc}\frac{dU_{dc}}{dt} = [u_{ia}i_a + u_{ib}i_b + u_{ic}i_c] - \frac{U_{dc}^2}{R_C} \end{cases} \tag{3}$$

Through Park transformation, the dynamic model of DSTATCOM can be derived out from (3), that is

$$\frac{d}{dt}\begin{bmatrix} i_d \\ i_q \\ U_{dc} \end{bmatrix} = \begin{bmatrix} -\frac{R_S}{L} & \omega & -\frac{\sqrt{3}K_T}{3L} m\cos\delta \\ -\omega & -\frac{R_S}{L} & -\frac{\sqrt{3}K_T}{3L} m\sin\delta \\ \frac{\sqrt{3}K_T}{2C} m\cos\delta & \frac{\sqrt{3}K_T}{2C} m\sin\delta & -\frac{1}{CR_C} \end{bmatrix} \begin{bmatrix} i_d \\ i_q \\ U_{dc} \end{bmatrix} + \begin{bmatrix} \frac{\sqrt{2}U_S}{L} \\ 0 \\ 0 \end{bmatrix} \tag{4}$$

IV. INTRODUCTION ABOUT INVERSE-SYSTEM METHOD

The basic method to design a multivariable nonlinear control system based on inverse-system[10] is as follows:

1) Deriving out the inverse system Π from the original system Σ.

2) Deriving out the corresponding α-order integral inverse-system Σ_α from Π.

3) Constructing a pseudo linear system Σ_l, which comprises Σ_α and Σ. Σ_l should be constructed to a simple equivalent form with feedback loop.

4) Considering the pseudo linear systemΣ_l with feedback loop as to be controlled object, therefore, we can design the controller based on linear system theory.

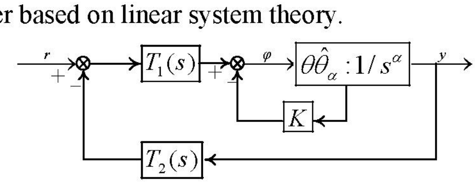


Fig. 4. The schematic diagram of designing a control system based on inverse-system method

The schematic diagram of designing a control system based on inverse-system is shown in Fig.4. Where $\theta\hat{\theta}_\alpha$ is the pseudo linear system to be controlled, $T_1(s)$ and $T_2(s)$ are physically realizable regulators, and K is state feedback coefficient. $T_1(s)$, $T_2(s)$ and K can be determined by different method in order to obtain the expected performance of the control system.

V. Reactive-voltage Decoupled PWM Control Based on Inverse-system Method

A. Model and State Reconstruction

For a dynamic model described by (4), we can not obtain the stable control law to realize the reactive-voltage decoupled design, if the inverse-system method is applied directly, since its inner zero-dynamic subsystem is unstable. Therefore, the first task we must do is to reconstruct model given in (4).

We choose the following variables as input variables

$$\boldsymbol{u} = [u_1 \quad u_2]^T = \left[\frac{\sqrt{3}}{3}K_T m\cos\delta \quad \frac{\sqrt{3}}{3}K_T m\sin\delta\right]^T \tag{5}$$

After Park transformation, the d-q components of DSTATCOM output voltage are

$$\begin{cases} V_d = \dfrac{\sqrt{3}}{3}K_T m U_{dc}\cos\delta = u_1 U_{dc} \\ V_q = \dfrac{\sqrt{3}}{3}K_T m U_{dc}\sin\delta = u_2 U_{dc} \end{cases} \tag{6}$$

Therefore, from (5) and (6), we have

$$u_1 = V_d / U_{dc}, u_2 = V_q / U_{dc} \tag{7}$$

Substituting (7) into (4), yields

$$\begin{cases} \dfrac{d}{dt}i_d = -\dfrac{R_S}{L}i_d + \omega i_q - \dfrac{1}{L}V_d + \dfrac{\sqrt{2}U_S}{L} \\ \dfrac{d}{dt}i_q = -\omega i_d - \dfrac{R_S}{L}i_q - \dfrac{1}{L}V_q \\ \dfrac{d}{dt}U_{dc} = -\dfrac{3}{2CU_{dc}}V_d i_d + \dfrac{3}{2CU_{dc}}V_q i_q - \dfrac{1}{CR_C}U_{dc} \end{cases} \tag{8}$$

Because the active power absorbed by DSTATCOM is:

$$P_{DSTATCOM} = \frac{3}{2}\left(V_d i_d + V_q i_q\right)$$

And the total input active power is:

$$P_{SYSTEM} = \frac{3}{2}E_d i_d = \frac{3\sqrt{2}}{2}U_S i_d$$

Suppose the active power transmission loss ratio is k, we have

$$P_{SYSTEM} = P_{DSTATCOM} + kP_{SYSTEM}$$

Therefore, the last equation in (8) can be rewritten as:

$$\frac{d}{dt}(U_{dc}) = \frac{3\sqrt{2}}{2CU_{dc}}(1-k)U_S i_d - \frac{1}{CR_C}U_{dc}$$

or

$$\frac{d}{dt}\left(U_{dc}^2\right) = 2U_{dc}\frac{d}{dt}U_{dc} = \frac{3\sqrt{2}}{C}(1-k)U_S i_d - \frac{2}{CR_C}U_{dc}^2$$

If we change one state variable from U_{dc} to U_{dc}^2, (8) becomes into

$$\begin{cases} \dfrac{d}{dt}i_d = -\dfrac{R_S}{L}i_d + \omega i_q - \dfrac{1}{L}V_d + \dfrac{\sqrt{2}U_S}{L} \\ \dfrac{d}{dt}i_q = -\omega i_d - \dfrac{R_S}{L}i_q - \dfrac{1}{L}V_q \\ \dfrac{d}{dt}\left(U_{dc}^2\right) = \dfrac{3\sqrt{2}}{C}(1-k)U_S i_d - \dfrac{2}{CR_C}U_{dc}^2 \end{cases} \tag{9}$$

Because of the uni-polar nature of the dc voltage, to consider U_{dc}^2 as the state variable does not cause a multi-solution problem.

B. Controller Design Based on Inverse-system Method

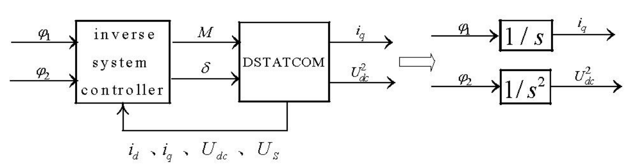
Based on the inverse-system method, letting i_q、U_{dc}^2 as the output variables, we can derive out the inverse-system control law of DSTATCOM from dynamic model described in (9), that is

$$\begin{bmatrix} V_d \\ V_q \end{bmatrix} = \begin{bmatrix} -R_S - \dfrac{2L}{CR_C} & \omega L & \dfrac{2\sqrt{2}L}{3CR_C^2(1-k)U_S} \\ -\omega L & -R_S & 0 \end{bmatrix}\begin{bmatrix} i_d \\ i_q \\ U_{dc}^2 \end{bmatrix} + \begin{bmatrix} \dfrac{\sqrt{2}CL}{6(1-k)U_S}\varphi_2 \\ -L\varphi_1 \end{bmatrix} + \begin{bmatrix} \sqrt{2}U_S \\ 0 \end{bmatrix} \tag{10}$$

And the inverse-system model Π is:

$$\begin{cases} m = \dfrac{\sqrt{\left[\left(-R_S - \dfrac{2L}{CR_C}\right)i_d + \omega L i_q + \dfrac{2\sqrt{2}L}{3CR_C^2(1-k)U_S}U_{dc}^2 - \dfrac{\sqrt{2}CL}{6(1-k)U_S}\varphi_2 + \sqrt{2}U_S\right]^2 + \left(-\omega L i_d - R_S i_q - L\varphi_1\right)^2}}{\dfrac{\sqrt{3}}{3}K_T U_{dc}} \\ \delta = arc\tan\left(\dfrac{-\omega L i_d - R_S i_q - L\varphi_1}{\left(-R_S - \dfrac{2L}{CR_C}\right)i_d + \omega L i_q + \dfrac{2\sqrt{2}L}{3CR_C^2(1-k)U_S}U_{dc}^2 - \dfrac{\sqrt{2}CL}{6(1-k)U_S}\varphi_2 + \sqrt{2}U_S}\right) \end{cases} \tag{11}$$

Considering the inverse-system model given in (11) as feedforward control of the original system(shown in Fig.5.(a)), then DSTATCOM and the feedforward control can be equivalent as a α = (1,2) order integral decoupled pseudo linear system, just as shown in Fig.5.(b).



(a) (b)

Fig. 5. The equivalent control diagram of DSTATCOM with inverse-system feedforward control

And $\alpha = (1,2)$ order decoupled pseudo linear system can be expressed as

$$\Sigma_i \begin{cases} \dot{\xi} = \varphi_2 \\ \begin{bmatrix} \dot{i}_d \\ \dot{i}_q \\ \dot{U}_{dc}^2 \end{bmatrix} = \begin{bmatrix} \dfrac{2}{CR_C} & 0 & -\dfrac{2\sqrt{2}}{3CR_C^2(1-k)U_S} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} i_d \\ i_q \\ U_{dc}^2 \end{bmatrix} \\ \quad + \begin{bmatrix} 0 & \dfrac{\sqrt{2}C}{6(1-k)U_S} \\ 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \varphi_1 \\ \varphi_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \xi \\ \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_d \\ i_q \\ U_{dc}^2 \end{bmatrix} \end{cases} \tag{12}$$

Where $\varphi \in R^2$ is the input vector, $y \in R^2$ is the output vector, $[\xi \quad x]^T \in R^{n+\dim(\xi)} = R^4$ is extending state vector. Obviously the decoupled pseudo linear system satisfies: $[\dot{y}_1 \quad \ddot{y}_2]^T = [\varphi_1 \quad \varphi_2]^T$, and its open-loop input/output transfer function is:

$$G(s) = diag\left(\frac{1}{s}, \frac{1}{s^2}\right) \tag{13}$$

The reactive-voltage decoupled control diagram of DSTATCOM is shown in Fig.6.

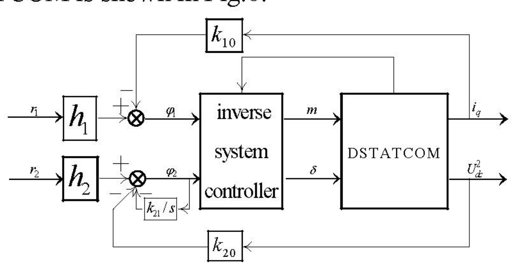


Fig. 6. The reactive-voltage decoupled control diagram of DSTATCOM

In order to control the reactive current and dc-link voltage effectively, we suppose the expected close-loop transfer function matrix is

$$W_d(s) = diag\left(\frac{h_1}{s+k_{10}}, \frac{h_2}{s^2+k_{21}s+k_{20}}\right) \tag{14}$$

The further feedback solution needed can be yielded as:

$$\varphi_1 = h_1 r_1 - k_{10} y_1 \tag{15}$$

$$\varphi_2 = h_2 r_2 - k_{20} y_2 - k_{21} \xi_1 \tag{16}$$

Where r_1, r_2 are reference inputs. The equivalent dynamic reactive-voltage diagram is shown in Fig.7.

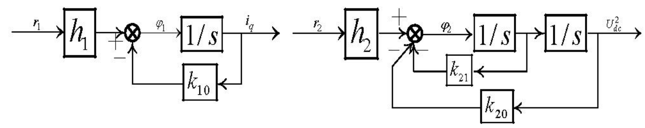


Fig. 7. The equivalent dynamic reactive-voltage diagram

C. Stability Analysis

We can infer from (12) that a 1-order zero-dynamic subsystem exist in the decoupled pseudo linear system (see also (17)), the stability of decoupled pseudo linear system can not be sure unless the zero-dynamic subsystem is stable.

$$\dot{i}_d = \frac{2}{CR_C} i_d - \frac{2\sqrt{2}}{3CR_C^2(1-k)U_S} U_{dc}^2 + \frac{\sqrt{2}C}{6(1-k)U_S} \varphi_2 \tag{17}$$

Substituting (16) into (17), yields:

$$si_d = -\frac{2}{CR_C} i_d - \frac{2\sqrt{2}}{3CR_C^2(1-k)U_S} U_{dc}^2$$
$$+ \frac{\sqrt{2}C}{6(1-k)U_S}(h_2 r_2 - k_{20} y_2 - k_{21} \xi_1) \tag{18}$$

On the other hand, from the close-loop transfer function described in (14), we have

$$U_{dc}^2 = \frac{h_2}{s^2 + k_{21}s + k_{20}} r_2 \tag{19}$$

From (18) and (19), transfer function $\dfrac{i_d}{r_2}$ can be obtained easily as

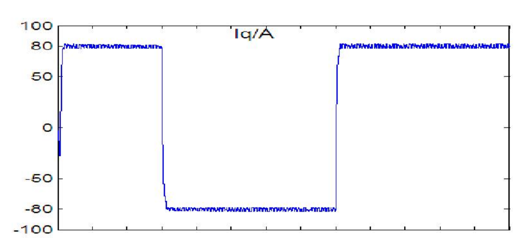
$$\frac{i_d}{r_2} = \frac{\sqrt{2}h_2(CR_C s + 2)}{6R_C(1-k)U_S(s^2 + k_{21}s + k_{20})} \tag{20}$$

We can observe that i_d and U_{dc}^2 have same poles, and the poles can be set arbitrarily--they don't depend on the operating point and circuit parameters. Therefore, the inner zero-dynamic subsystem is fully stable, the stability of the total system can be assured by the designed control law.

VI. Simulation Results

The simulation analyses on inverse-system reactive current-voltage decoupled control of DSTATCOM are implemented based on MATLAB/SIMULINK software.

The reference reactive current changes suddenly from 80A to -80A at t=0.06s, and it return to 80A at t=0.16s. the simulation results are shown as Fig.8.



(a)

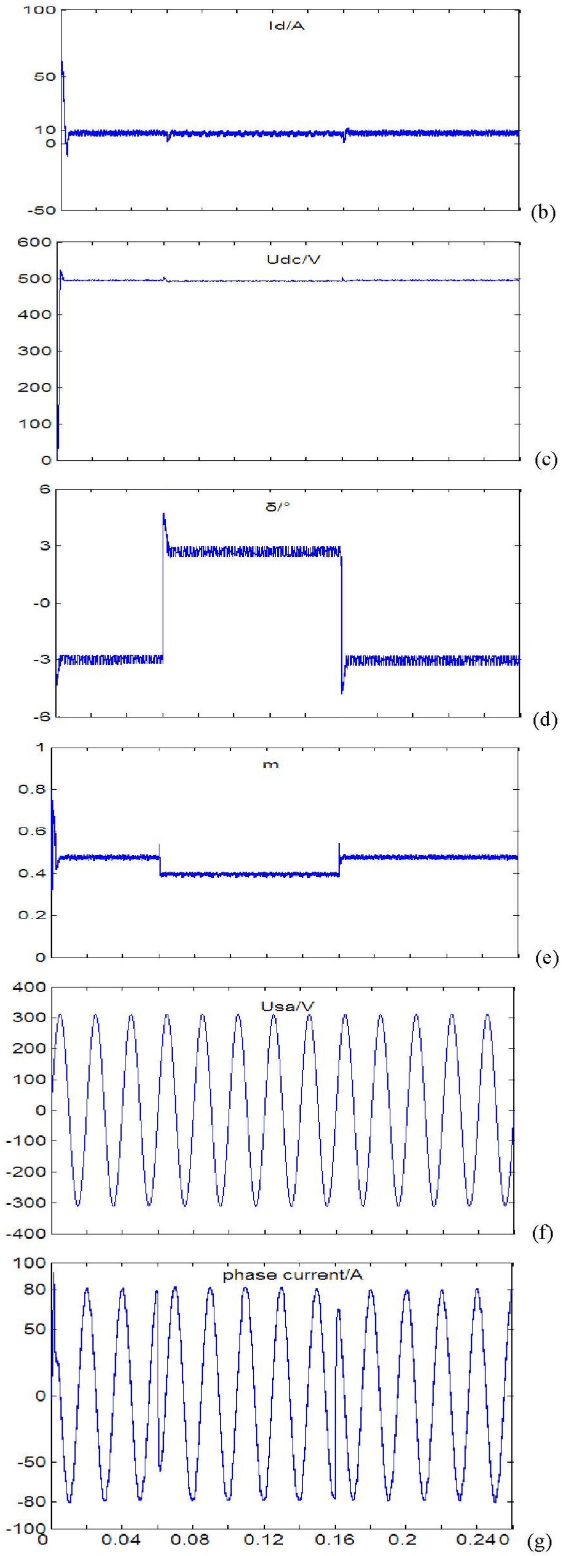


Fig. 8. The step responses when the reference reactive current changes in step. (a).i_q, (b).i_d, (c) U_{dc}, (d) δ, (e) m, (f)System voltage,(g)Phase current

The simulation results show that when the reactive-voltage decoupled PWM control based on inverse-system are employed, fast responses of reactive current and dc-link voltage within 5ms are achieved, the reactive current response is accurate, and the voltage response has a transitory approximate 5% overshot(within 2ms).

VII. Conclusions

The mathematical model of a DSTATCOM is first established at the basis of choosing its main circuit configuration in this paper, then a reactive-voltage decoupled PWM controller is designed based on multivariable inverse-system method, the controller can assure the full stability of the total system. By this method, the controller parameters' design is simplified effectively. The simulation results show that a fast response of reactive current and dc-link voltage within 5ms is achieved using the proposed dynamic control method, and the dynamic voltage stability of the device has been increased effectively.

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IX. Biographies

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